+61 406203225
Sydney, NSW

Murtaza Pakawala

mypakawala@hotmail.com
Github

Education

Sydney, Australia | **UNSW** | **Feb 2022 - Dec 2025**

- Bachelor of Advanced Computer Science (Honours) with a Minor in Mathematics.
- Engineering WAM: 80.62
- Coursework: Machine Learning and Data Mining, Computer Vision, DeepLearning, Advance C++, Programming Challenges, Extended Operating Systems

Ann Arbor, USA | **University Of Michigan** | **Jan 2025 - May 2025**

- Exchange Student
- Coursework: Linear Algebra, Number theory, Probablity

Research Thesis

UNSW Sydney | **May 2025 – Present**

- **Thesis:** Inferring Evolutionary Parameters from Phylogenies.
- Developing a simulation-based inference pipeline using **msprime** and **PyTorch** to estimate evolutionary parameters from phylogenetic data.
- Benchmarking novel deep learning approaches (**NPE** and **FMPE**) against standard **Approximate Bayesian Computation (ABC)** methods.

Personal Projects

- **C++ Neural Network from Scratch** *GitHub Repository* Designed and implemented a neural network from scratch in **C++**, leveraging object-oriented programming (OOP) principles to modularize different components. Implemented gradient computation and backpropagation using the chain rule, providing a hands-on understanding of the inner workings of neural networks.
- **Course Tracker** *GitHub Repository*
  Designed and developed a **React Native** application to help university students track course progress and gain insights into their overall and individual marks. Built with **Express.js**, **TypeScript**, and **SQL**, the app won 2nd place overall in the Personal Project Competition and was voted the best app by UNSW students.

Extracurriculars

CSESoc Projects

GitHub Repository

- Contributed to the Discord Bot project used by thousands of students in the CSESoc Discord server.
- Successfully migrated the codebase from JavaScript (JS) to TypeScript (TS), improving maintainability and enforcing strong typing.
- Developed a course join/leave feature, allowing students to easily join course-specific chat communities.

Skills

- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
- **Programming Languages:** C++, C, Java, Python, JavaScript, TypeScript, SQL
- **Frameworks & Technologies:** React, React Native, Next.js, Express.js, NestJS, PostgreSQL

Awards

- Second Prize in a personnel project competition organized by CSESOC
- Highest Voted App prize organized by CSESOC
- Yale Young Global Scholar Award (YYGS)-2020